

CISCO NETWORKING PROJECT

Project 1: Static Routing

Multi-Router Static Route Configuration and Verification using Cisco Packet Tracer

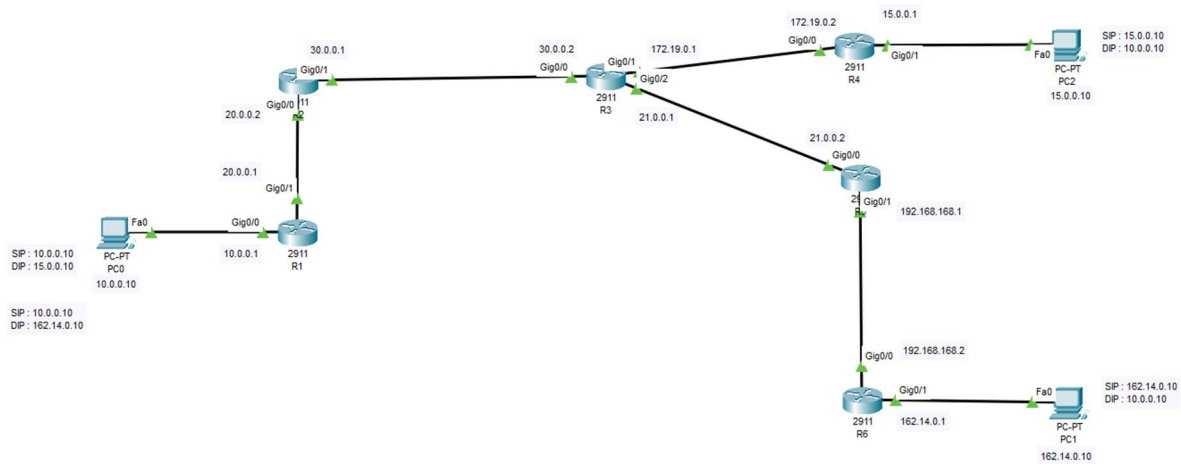


Figure — Full Network Topology (6 Routers, 3 Hosts)

Routers	R1 – R6 (Cisco 2911)
Routing Method	Static Routes
End Devices	PC0, PC1, PC2
Simulation Tool	Cisco Packet Tracer

1. Project Objective

This project demonstrates end-to-end IP reachability across a six-router Cisco network using static routing. The network consists of eight distinct network IDs, and the goal is to ensure every router holds a complete route to all eight networks so that traffic can be exchanged between the three end hosts (PC0, PC1, and PC2).

The task involves the following stages, in order:

- Record all IP networks shown in the topology and assign interface addressing to each router.
- Bring up directly connected routes by activating each router interface.
- Verify reachability router-by-router and identify any missing routes.
- Configure static routes to fill the gaps and confirm end-to-end connectivity with ping tests.

1.1 Network IDs in the Topology

A total of eight networks must be present in every router's routing table for full reachability:

1. 10.0.0.0 /8	2. 15.0.0.0 /8
3. 20.0.0.0 /8	4. 21.0.0.0 /8
5. 30.0.0.0 /8	6. 162.0.0.0 /8
7. 172.0.0.0 /8 (172.19.0.0 /16 used)	8. 192.0.0.0 /8 (192.168.168.0 /24 used)

2. Network Topology

The topology consists of six routers connected in a linear/branching backbone, with end-user hosts attached at the far edges. PC0 (10.0.0.10) sits behind R1; PC2 (15.0.0.10) sits behind R4; PC1 (162.14.0.10) sits behind R6.

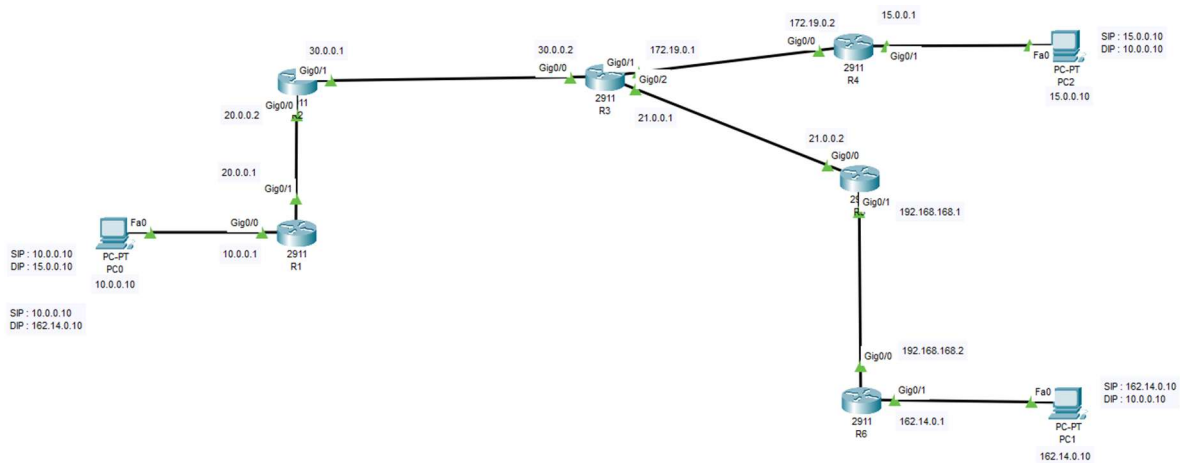


Figure 1 — Network topology as built in Cisco Packet Tracer

3. IP Addressing Plan

Once the networks were identified, each router interface was assigned an IP address matching the topology diagram. Assigning the interface addresses automatically creates the corresponding directly-connected routes in each router's routing table.

3.1 Router Interface Addressing

Router	Interface	IP Address	Prefix	Purpose
R1	GigabitEthernet0/0	10.0.0.1	/8	LAN link to PC0
R1	GigabitEthernet0/1	20.0.0.1	/8	Link to R2
R2	GigabitEthernet0/0	20.0.0.2	/8	Link to R1
R2	GigabitEthernet0/1	30.0.0.1	/8	Link to R3
R3	GigabitEthernet0/0	30.0.0.2	/8	Link to R2
R3	GigabitEthernet0/1	172.19.0.1	/16	Link to R4
R3	GigabitEthernet0/2	21.0.0.1	/8	Link to R5
R4	GigabitEthernet0/0	172.19.0.2	/16	Link to R3
R4	GigabitEthernet0/1	15.0.0.1	/8	LAN link to PC2
R5	GigabitEthernet0/0	21.0.0.2	/8	Link to R3
R5	GigabitEthernet0/1	192.168.168.1	/24	Link to R6
R6	GigabitEthernet0/0	192.168.168.2	/24	Link to R5
R6	GigabitEthernet0/1	162.14.0.1	/16	LAN link to PC1

3.2 End-Device (PC) Addressing

Host	IP Address	Subnet Mask	Default Gateway	Connected To
PC0	10.0.0.10	255.0.0.0	10.0.0.1	R1 Gig0/0 LAN
PC2	15.0.0.10	255.0.0.0	15.0.0.1	R4 Gig0/1 LAN
PC1	162.14.0.10	255.255.0.0	162.14.0.1	R6 Gig0/1 LAN

4. Baseline Interface Configuration

Before configuration, all router interfaces are administratively down with no IP address assigned, as shown by show ip interface brief on R1:

```
R1#show ip int bri
R1#show ip int brief
Interface                IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0       unassigned      YES unset    administratively down down
GigabitEthernet0/1       unassigned      YES unset    administratively down down
GigabitEthernet0/2       unassigned      YES unset    administratively down down
Vlan1                    unassigned      YES unset    administratively down down
R1#
```

Figure 2 — show ip interface brief before configuration (R1)

Each interface was then configured with its IP address and enabled with no shutdown. Example configuration on R1:

```
R1#configure terminal
R1(config)#interface gig0/0
R1(config-if)#ip address 10.0.0.1 255.0.0.0
R1(config-if)#no shutdown
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#interface gig0/1
R1(config-if)#ip address 20.0.0.1 255.0.0.0
R1(config-if)#no shutdown
```

```

R1#conf t
R1#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int gig0/0
R1(config-if)#ip add 10.0.0.1 255.0.0.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#int gig0/1
R1(config-if)#ip add 20.0.0.1 255.0.0.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

R1(config-if)#exit
R1(config)#

```

Figure 3 — Interface configuration commands entered on R1

The same ip address / no shutdown sequence was repeated on every interface of R2, R3, R4, R5 and R6 using the addressing plan in Section 3.

5. Verifying Directly Connected Routes

After configuring R1 only, Gig0/1 shows protocol down, because the neighboring router (R2) has not yet been configured — there is no device on the other end of the link to complete the up/up state.

```

R1#show ip int bri
R1#show ip int brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.0.0.1	YES	manual	up	up
GigabitEthernet0/1	20.0.0.1	YES	manual	up	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```

R1#

```

Figure 4 — show ip interface brief: Gig0/1 protocol down before R2 is configured

Consequently, the 20.0.0.0/8 network is missing from R1's routing table, because a Cisco router only installs a connected route once the interface line protocol is up.

```

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0

R1#

```

Figure 5 — show ip route: 20.0.0.0 network absent until the R2 link comes up

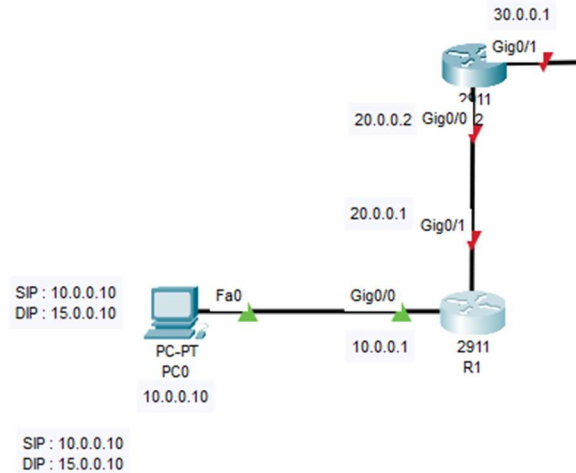


Figure 6 — Detail: link between R1 and R2 shown down in the topology

Once R2's matching interface (Gig0/0, 20.0.0.2) is configured and enabled, the link comes up and the 20.0.0.0/8 network appears automatically in R1's route table as a directly connected (C) route.

```
R2#conf t
R2#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int gig0/0
R2(config-if)#ip add 20.0.0.2 255.0.0.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2(config-if)#exit
R2(config)#
```

Figure 7 — R2 interfaces after configuration

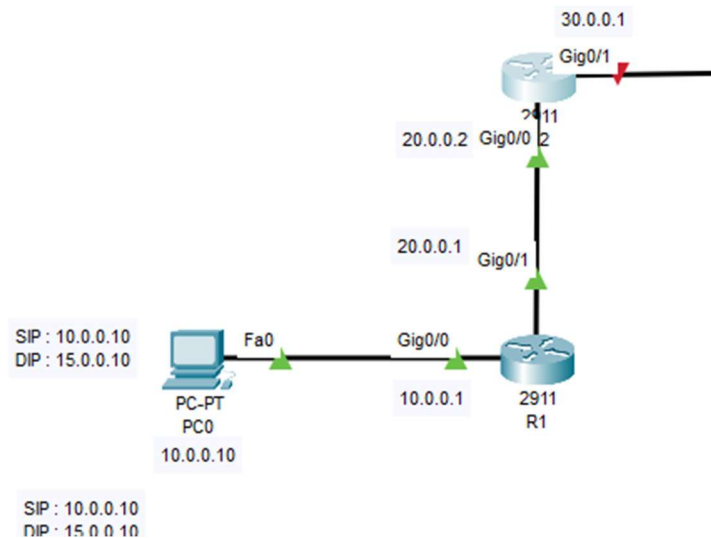


Figure 8 — Link state up between R1 and R2

R1's routing table now shows the 20.0.0.0/8 network automatically, confirming the directly connected route was installed:

```

R1>enable
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, GigabitEthernet0/1
L       20.0.0.1/32 is directly connected, GigabitEthernet0/1

R1#

```

Figure 9 — show ip route on R1: 20.0.0.0/8 now present

This same process — assign IP address, enable the interface, verify the link and the connected route — was repeated for every remaining interface on R2, R3, R4, R5 and R6 until all directly connected networks matched the addressing plan in Section 3.

6. Identifying Missing (Non-Connected) Routes

With all interfaces up, each router only knows about the networks it is directly connected to. Networks that are more than one hop away are still missing and must be reached with static routes. End devices were assigned IP addressing on their NICs matching the plan in Section 3.2, and a ping test was run between PC0 and PC2 to confirm whether reachability already exists.

Testing connectivity between PC0 (10.0.0.10) and PC2 (15.0.0.10) fails, since none of the intermediate routers has a route to the remote network yet — this confirms static routes are required.

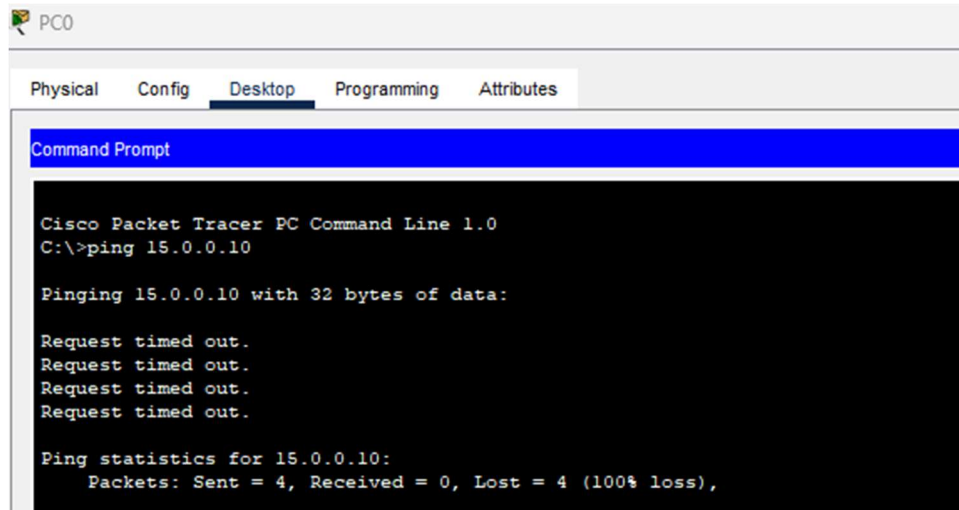


Figure 10 — Ping from PC0 (10.0.0.10) to PC2 (15.0.0.10) times out before static routes exist

Path 10.0.0.10 → 15.0.0.10 requires static routes on R1, R2 and R3 pointing toward the 15.0.0.0/8 network (R4 is directly connected to it). The return path 15.0.0.10 → 10.0.0.10 requires static routes on R4, R3 and R2 pointing back toward 10.0.0.0/8 (R1 is directly connected to it).

7. Static Route Configuration

Static routes were added with the ip route command in the form: ip route <destination network> <subnet mask> <next-hop IP>.

7.1 Path: PC0 (10.0.0.10) ↔ PC2 (15.0.0.10)

Forward direction — routing toward 15.0.0.0/8

Router	Configuration
R1	ip route 15.0.0.0 255.0.0.0 20.0.0.2
R2	ip route 15.0.0.0 255.0.0.0 30.0.0.2
R3	ip route 15.0.0.0 255.0.0.0 172.19.0.2
R4	Directly connected to 15.0.0.0/8 — no static route needed

The command below shows the route being added on R1, and the resulting routing table with the new static (S) entry for 15.0.0.0/8 via 20.0.0.2:

```
R1#conf t
R1#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip route 15.0.0.0 255.0.0.0 20.0.0.2
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
S       15.0.0.0/8 [1/0] via 20.0.0.2
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, GigabitEthernet0/1
L       20.0.0.1/32 is directly connected, GigabitEthernet0/1
```

Figure 11 — R1: ip route 15.0.0.0 255.0.0.0 20.0.0.2 configured; show ip route confirms the new S route

Return direction — routing toward 10.0.0.0/8

Router	Configuration
R4	ip route 10.0.0.0 255.0.0.0 172.19.0.1
R3	ip route 10.0.0.0 255.0.0.0 30.0.0.1
R2	ip route 10.0.0.0 255.0.0.0 20.0.0.1
R1	Directly connected to 10.0.0.0/8 — no static route needed

Before the return-path routes are added, PC2 can already reach as far as R4, but the reply is rejected further upstream ('Destination host unreachable' from 15.0.0.1) because R4 does not yet know how to route back to 10.0.0.0/8:

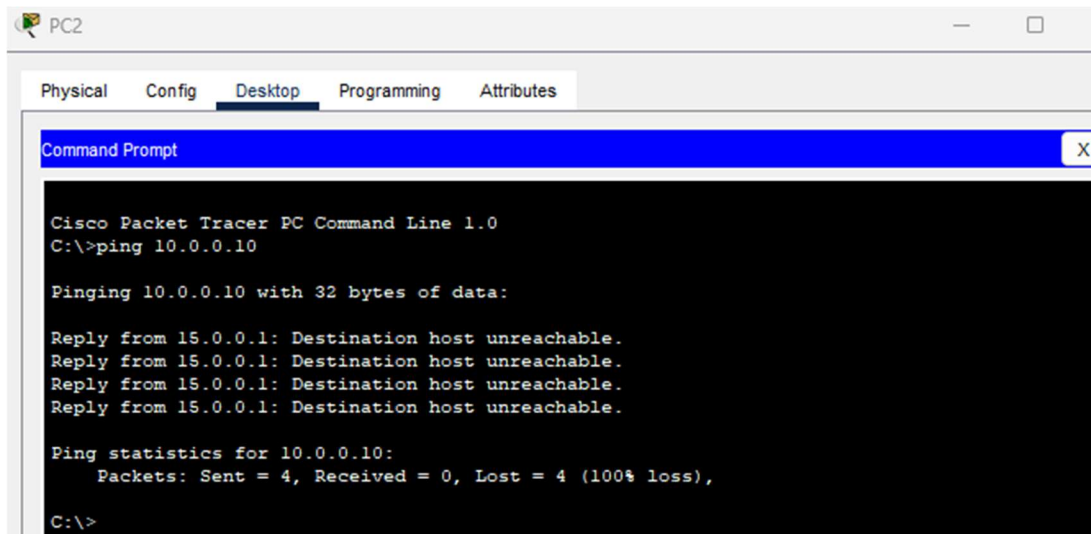


Figure 12 — Return path from PC2 (15.0.0.10) still failing until R4, R3 and R2 get their static route back to 10.0.0.0/8

Once the return-path static routes above are added, verification confirms full success — ping from PC0 to PC2 now succeeds:

```
C:\>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time<1ms TTL=124
Reply from 15.0.0.10: bytes=32 time<1ms TTL=124
Reply from 15.0.0.10: bytes=32 time<1ms TTL=124
Reply from 15.0.0.10: bytes=32 time<1ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 13 — Successful ping: 10.0.0.10 → 15.0.0.10 (0% loss)

7.2 Path: PC0 (10.0.0.10) ↔ PC1 (162.14.0.10)

Forward direction — routing toward 162.0.0.0/8

Router	Configuration
R1	ip route 162.0.0.0 255.0.0.0 20.0.0.2
R2	ip route 162.0.0.0 255.0.0.0 30.0.0.2
R3	ip route 162.0.0.0 255.0.0.0 21.0.0.2
R5	ip route 162.0.0.0 255.0.0.0 192.168.168.2
R6	Directly connected to 162.0.0.0/8 — no static route needed

Return direction — routing toward 10.0.0.0/8

Router	Configuration
R6	ip route 10.0.0.0 255.0.0.0 192.168.168.1

Router	Configuration
R5	ip route 10.0.0.0 255.0.0.0 21.0.0.1
R3	Route to 10.0.0.0/8 already exists (added in Section 7.1)
R2	Route to 10.0.0.0/8 already exists (added in Section 7.1)
R1	Directly connected to 10.0.0.0/8 — no static route needed

With all static routes in place, every host can now reach every other host across the topology:

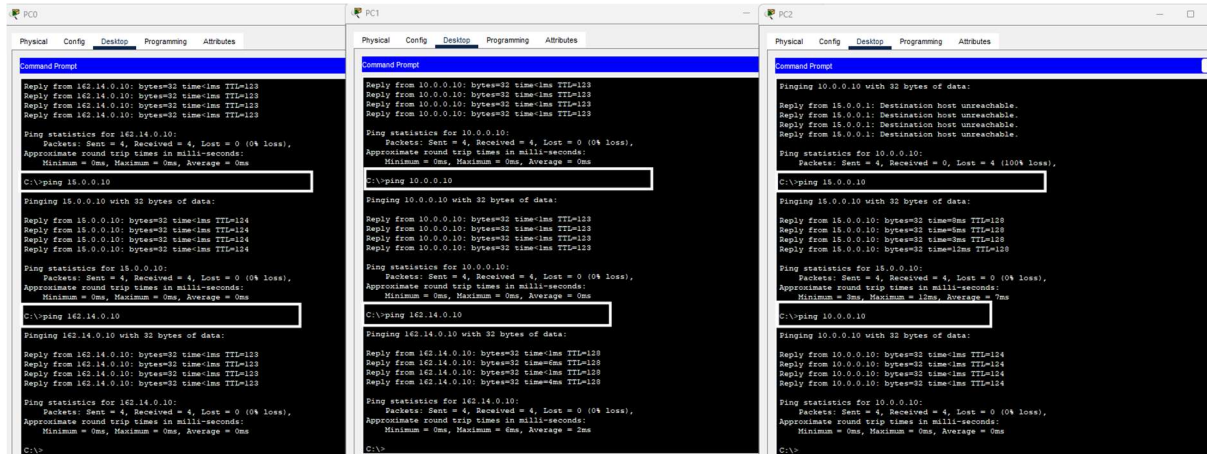


Figure 14 — Full topology after configuration: all links up, static routes installed, end-to-end ping successful

8. End-to-End Path Summary

8.1 PC0 (10.0.0.10) → PC2 (15.0.0.10)

PC0 → R1 → R2 → R3 → R4 → PC2

8.2 PC0 (10.0.0.10) → PC1 (162.14.0.10)

PC0 → R1 → R2 → R3 → R5 → R6 → PC1

9. Conclusion

All eight required networks were successfully installed into the routing table of every router, using a combination of directly connected routes (from interface addressing) and manually configured static routes. Ping tests confirmed full, bidirectional reachability between PC0, PC1 and PC2 across all six routers, meeting the project objective.

Key takeaways from this exercise:

- A directly connected route only appears once both ends of a link are configured and the line protocol is up.

- Static routes must be configured in both directions — the forward path and the return path each need their own route on every intermediate router.
- A router that already carries a route to a network from an earlier task (e.g. R2/R3 already routing to 10.0.0.0/8) does not need that route re-added for a new destination.
- `show ip interface brief` and `show ip route` are the primary tools for diagnosing missing connectivity before and after static route configuration.